Question 19

Aim

To solve the recurrence relation $T(n) = T(n/2) + n$ using the Substitution Method and determine its time complexity.

Procedure

1. Start the program.
2. Read the input size (n) from the user.
3. Identify the recurrence relation $T(n) = T(n/2) + n$.
4. Expand the recurrence relation step-by-step using the Substitution Method.
5. Continue the expansion until the base case is reached.
6. Simplify the expanded recurrence.
7. Determine the time complexity.
8. Display the result.
9. Stop the program.

Algorithm

1. Start.
2. Input the value of n.
3. If $n = 1$, return 1.
4. Otherwise, recursively compute $T(n/2)$.
5. Add the current value of n to the recursive result.
6. Repeat until the base case is reached.
7. Display the computed result.
8. Display the time complexity $\Theta(n)$.
9. Stop.

Input

Enter input size (power of 2): 16

Output

Result = 31

Time Complexity = $\Theta(n)$

Result

The recurrence relation $T(n) = T(n/2) + n$ was solved using the Substitution Method. The algorithm reduces the input size by half at each recursive call and has a final time complexity of $\Theta(n)$.